

rates, the dollar against the yen, the dollar against the European currencies, the European stock markets, the United States balance of payments, United States inflation, and another dozen prime factors, then the journalists need to look at all of these factors, look at their historical effect both in isolation and jointly, look at the stability of such influence, then, after consulting the test statistic isolate the factor if it is possible to do so. Finally, a proper confidence level needs to be given to the factor itself; if it is less than 90% the story would be dead. I can understand why Hume was extremely obsessed with causality and could not accept such inference anywhere.

I have a trick to know if something *real* in the world is taking place. I have set up my Bloomberg monitor to display the price and percentage change of all relevant prices in the world: currencies, stocks, interest rates, and commodities. By dint of looking at the same setup for years, as I keep the currencies in the upper left corner and the various stock markets on the right, I managed to build an instinctive way of knowing if something serious is going on. The trick is to look only at the large percentage changes. Unless something moves by more than its usual daily percentage change, the event is deemed to be noise. Percentage moves are the size of the headlines. In addition, the interpretation is not linear; a 2% move is not twice as significant an event as 1%, it is rather like four times. The headline of the Dow moving by 1.3 points on my screen today has less than one millionth of the significance of the serious 7% drop of October 1997. People might ask me: why do I want everybody to learn some statistics? The answer is that too many people read explanations. We cannot instinctively understand the nonlinear aspect of probability.

FILTERING METHODS

Engineers use methods to clean up the noise from the signal in the data. Did it ever occur to you while talking to your cousin in Australia or the South Pole that the static on the telephone line could be distinguished from the voice of your correspondent? The method is to consider that when a change in amplitude is small, it is more likely to result from noise – with its likelihood of being a signal increasing exponentially as

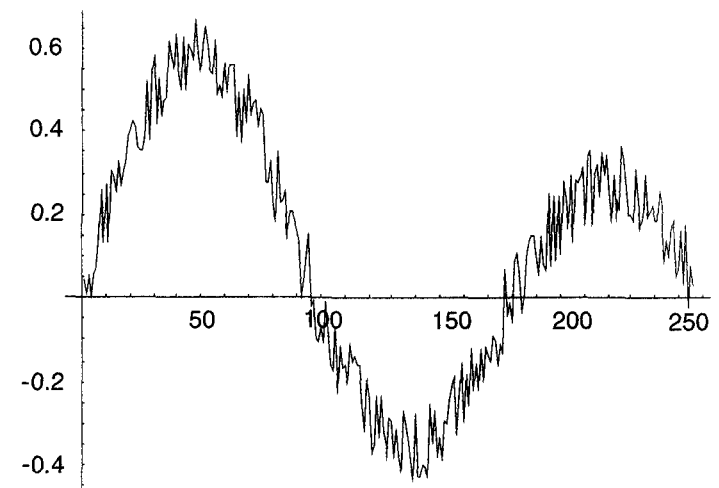


Figure 11.1 Unfiltered Data Containing Signal and Noise.

its magnitude increases. The method is called a smoothing kernel which has been applied in Figures 11.1 and 11.2. But our auditory system is incapable of performing such a function by itself. Likewise our brain cannot see the difference between a significant price change and mere noise, particularly when it is pounded with unsmoothed journalistic noise.

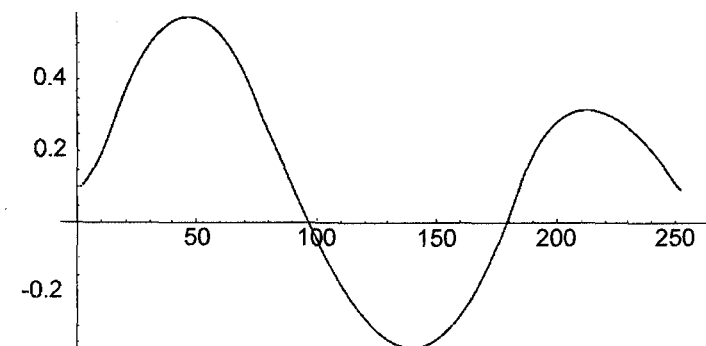


Figure 11.2 Same Data With its Noise Removed.